

seamlessly. One can opt for part-MATLAB, part-COMSOL codes. A MATLAB user will not see the difference, but the same type of commands.

How can a user try simulation using a proximity sensor or radar?

The exact operation of the sensors can be simulated and users can then use COMSOL to define the dormitory of the sensors. If it is defined in CAD software, one can have the exact representation of the sensor dormitory. COMSOL is used to simulate the radar propagation from the sensor and how it bounces back from a wall. It can correlate between bounce-back and detection too.

COMSOL multiphysics is a base platform. There is an add-on product called the RF module dedicated to radio radar and microwave type of applications. It is used for communication too, i.e., for understanding the electromagnetic wave propagation as well as in optical waves.

What are the offerings from COMSOL for an electronics enthusiast?

We have several offerings for electrical and electronics engineers depending on their specific area of interest. The AC-DC module is for low-frequency electrical applications such as electromagnetic interference/compatibility, modeling electrical motors and generators and circuit modeling, to name a few. The RF module is useful for a variety of

high-frequency applications such as photonics, microwaves and terahertz waves. Also, there is an MEMS module, useful while designing sensors, actuators and other such micron-sized devices.

In thermoelectric coolers for chips, thermoelectric effect is used. It is the same kind of phenomena as used in a portable refrigerator. There are peltier elements that drive the heat to one side thereby cooling the other side. COMSOL can simulate the process and show how it works. It is used to design the cooling packages for chips that are set directly on the chip. It does not require a fan, but drags the heat out by electrically-driven cooling. Thermoelectric cooling is an important aspect useful to electronics engineers.

Moreover, if solders on a chip heat up, these tend to become elastic and eventually break due to some structural intent. This process is known as structural creep. In the latest versions, there are a number of tools including elastoplastic creep type simulations that can help users predict the lifetime of a chip in terms of the solders, depending on the thermal load. When all this is combined together, a measured representative current input is received, which produces the heat that, in turn, makes the solder creep so it starts to flow.

By running these simulations (not in real time but afterwards), the life span of electronic components can be estimated by combining electrical, structural and heat transfer simulations. ●



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